Polar Coordinates and Conversions

ANSWER KEY

Converting points

1 Convert each polar point (r, θ) to rectangular coordinates (exact values):

a $\left(4, \frac{\pi}{3}\right)$ $(2, 2\sqrt{3})$

b $\left(2, \frac{3\pi}{4}\right)$ $(-\sqrt{2}, \sqrt{2})$

2 Convert each rectangular point to polar coordinates with $r \geq 0$ and $0 \leq \theta < 2\pi$:

a $(-3, 3)$ $\left(3\sqrt{2}, \frac{3\pi}{4}\right)$

b $(0, -5)$ $\left(5, \frac{3\pi}{2}\right)$

Converting equations

3 Convert the polar equation $r = 6\cos\theta$ to rectangular form and identify the curve.

Multiply by r: $r^2 = 6r\cos\theta$, so $x^2 + y^2 = 6x$, i.e. $(x-3)^2 + y^2 = 9$: a circle with center $(3, 0)$ and radius 3.

4 Convert each rectangular equation to polar form:

a $x^2 + y^2 = 16$ $r = 4$

b $y = x$ $\theta = \frac{\pi}{4}$

5 Convert the polar equation $r = \frac{2}{1 - \sin\theta}$ to rectangular form.

$r - r\sin\theta = 2$, so $r = y + 2$. Squaring: $x^2 + y^2 = (y+2)^2 = y^2 + 4y + 4$, so $x^2 = 4y + 4$ — a parabola.

Multiple representations

6 Give two other pairs (r, θ) that represent the same point as $\left(3, \frac{\pi}{4}\right)$: one with $r > 0$ and θ outside $[0, 2\pi)$, and one with $r < 0$.

Adding a full rotation: $\left(3, \frac{\pi}{4} + 2\pi\right) = \left(3, \frac{9\pi}{4}\right)$. Using negative r (add π to the angle): $\left(-3, \frac{\pi}{4} + \pi\right) = \left(-3, \frac{5\pi}{4}\right)$.

7 Plot-reasoning: explain why the polar point $\left(-2, \frac{\pi}{2}\right)$ corresponds to the same location as $\left(2, \frac{3\pi}{2}\right)$.

A negative r means the point lies in the direction opposite to θ. Pointing at angle $\frac{\pi}{2}$ (straight up) with $r = -2$ places the point 2 units in the opposite direction, straight down — the same location as $r = 2$ at angle $\frac{3\pi}{2}$ (also straight down).

Distance between polar points

8 Convert $\left(4, \frac{\pi}{6}\right)$ and $\left(4, \frac{2\pi}{3}\right)$ to rectangular coordinates, then find the distance between them, to two decimal places.

$\left(4, \frac{\pi}{6}\right) \to (2\sqrt{3}, 2)$; $\left(4, \frac{2\pi}{3}\right) \to (-2, 2\sqrt{3})$. Distance $= \sqrt{(2\sqrt{3}+2)^2 + (2-2\sqrt{3})^2} \approx \sqrt{29.86 + 2.14} = \sqrt{32} \approx 5.66$.

Further conversion practice

9 Convert each polar point to rectangular coordinates (exact values):

a $(6, \pi)$ $(-6, 0)$

b $\left(5, \frac{5\pi}{6}\right)$ $\left(-\frac{5\sqrt{3}}{2}, \frac{5}{2}\right)$

10 Convert the polar equation $r^2 = 4\sin 2\theta$ to note its symmetry, and identify it as a standard curve type.

This is a lemniscate (figure-eight curve). Rewriting $r^2 = 4\sin 2\theta = 8\sin\theta\cos\theta$, and using $r^2 = x^2 + y^2$, $r\cos\theta = x$, $r\sin\theta = y$: multiplying both sides by r^2 gives $(x^2 + y^2)^2 = 8xy$.